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Key Personnel Changes/Previous PI:	December 2015 report: Postdoc: Lin Zheng, PhD. Now professor of Mechanical Engineering in China ; Undergraduate: Mr. Omar Duran (minority student), 4th year BS candidate in Chemical Engineering - graduated 2013. PhD student and later postdoc: Dr. Jinyong Bao, PhD. Chem. Eng. August 2011, The motion of the three-phase contact line and its role in heat transfer in the growth of a vapor bubble on a heated solid surface.		
COI Name (Institution):	Lee, Taehun (City College of the City University of New York) Koplik, Joel (City College of the City University of New York)		
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Task Description:	High latent heats allow evaporating small amounts of fluid to cool hot surfaces very efficiently. Boiling is thus attractive for cooling nuclear reactors, computer chips, hot surfaces in space (where volume is at a premium), etc., provided the hot surface remains wet, which can be an issue in terrestrial and certainly is in microgravity. Dryout can be catastrophic. Despite decades of study, competing mechanisms for boiling heat transfer exist, including fluid mixing and liquid microlayer (under a portion of the vapor bubble) evaporation. We have carried out preliminary quasistatic heat conduction calculations of a single vapor bubble on the surface of a thin solid layer covered by a liquid of finite thickness. The solid sits on a heater at fixed temperature above the fluid's saturation temperature. The system is axisymmetric with periodic boundary conditions at its lateral edge to mimic bubble-bubble interactions. These conduction-only results match the pronounced heat transfer jump upon (surface) vapor bubble nucleation in all standard boiling curves and provide scalings of the heat flux with many key physical parameters that agree with historic correlations. Its simple mechanism may be transformative: The bubble's 3-phase contact line (CL) locally cools the superheated solid surface from an otherwise superheated value to the fluid saturation temperature. The contact line thus acts as a heat magnet from the solid for, e.g., microlayer vaporization. Critical questions include: How does the heat transfer vary as bubbles grow, deform, and detach? Does a detaching bubble leave behind a vapor residue/CL, generating a new bubble to draw heat? Does it detach cleanly? Or, does cooler fluid that replaces the rising bubble condense the residue bubble? A dynamic calculation addressing these questions needs CL motion with bubble growth as an input. Residue bubble quenching involves convection/convective transfer. We shall: 1. Create a molecular-sized model of our solid-liquid-vapor bubble system and let interparticle physics via molecular dynamics (MD) determine CL motion. We calculate bubble growth and heat transfer for a nanobubble directly from the MD results or input its CL motion into our quasistatic heat conduction model. This MD will begin with zero gravity and later introduce a pseudogravity so as to mimic a real bubble's surface tension-bouyancy balance (Bond number). 2. Formulate the fluid mechanics /heat transfer /phase change as a coarse-grained lattice-Boltzmann (LB) problem, which treats the liquid, its vapor and their mutual interface (including the CL) as a single fluid with a diffuse interface. One specifies the fluid-solid interaction energy; CL motion is part of the solution. This avoids the need to explicitly track the vapor-liquid interface and apply interfacial conditions there. In the presence of gravity, we use LB to follow deforming bubbles towards detachment for different system parameters, examine if a vapor residue remains on the surface after detachment and, if so, if convection quenches it. In gravity's absence we shall examine if the circulation provided by the expanding bubble's fluid displacement can help remove a bubble from the surface. We shall introduce intermittent fluid circulation from the bulk toward the surface along the domain boundary to push bubbles off and to convect them away from the surface. The goal is to avoid surface dryout, common in microgravity boiling, and to permit sustained, efficient surface cooling. If the calculations suggest this as being feasible, obvious microgravity/shuttle experiments would test it. In this and prior years we have extended our continuum calculation to describe the growth of individual bubbles on the surface and their detachment due to gravity, and developed a simple analytic theory to explain our findings. We have also begun to consider a constant heat flux, rather than a constant temperature heater condition and are examining the differences that it makes. We have completed a number of studies using LB and MD methods. See previous annual Task Book reports for details. In the last period, we examined the transition to film boiling, the situation where vapor takes over the entire solid surface momentarily, which could be catastrophic to sensitive electronics. We are using MD for this study.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A vastly improved understanding of the mechanism of nucleate boiling heat transfer and the role played by gravity in it can have significant impacts on both terrestrial life and that on humans in space. An advance that makes nucleate boiling heat transfer essentially risk-free would enable the use of far less fluid to cool hot surfaces. For example, it could lead to the use of far less water in the secondary cooling of nuclear reactors, and therefore lead to less thermal pollution of our waters. Instead of releasing large amounts of hot water into rivers, one might be able to release far less hot water and more steam. Such an understanding might also influence computer design, since cooling issues are a critical constraint on computer design. In space applications, such cooling without the high risk of burnout could lead to far smaller cooling systems - and since the total space in the space station or in a space ship is extremely limited, such an advance could be highly beneficial.
Task Progress:	Over the grant period, our team (mainly one PhD student, two postdocs, one PI, two co-Is) has taken a three-pronged approach to this problem. We have 1. Constructed and refined a continuum model of heat transfer from a solid heated from below to a vapor bubble in a liquid layer on its surface, with both solid and liquid layers of finite thickness, including both quasistatic temperature and heat flux profiles and their consequences as well as the quasistatic growth, necking and detachment of the vapor bubble. The latter calculation required the postulation of various ad hoc models for how the CL moves as the bubble grows. 2. A lattice-Boltzmann model of the composite system comprised of all three phases, which considers the liquid and vapor phases as a single phase of variable density and thus avoids the specification as an input of CL motion. 3. A molecular dynamics approach that simulates on the nanoscale the three-phase experiment, beginning with a liquid layer above a solid, both at uniform temperature below the liquid saturation temperature, through heating of the bottom of the solid, bubble nucleation, growth and, due to the action of a uniform body force, detachment. This simulation implicitly includes flow/convective heat transfer, albeit on the nanoscale. Portions of this work have been either published or are currently in review and others are in various stages of preparation (see below). The annual progress reports provided extensive details of portions of this work, including numerous figures. We shall give a brief overview of the highlights of the results. 1. The Continuum calculation: a. Quasistatic heat transfer to a vapor bubble in a liquid layer on a heating surface of finite thickness: Exploring the mechanism of nucleate boiling heat transfer-Part I. To investigate the problem of nucleate pool boiling we solve the quasistatic temperature and heat flux fields of a periodic system comprised of a vapor bubble on or near a solid-liquid interface, where both the solid and liquid layers are of finite thickness and where a heater maintains the bottom surface of the solid at a constant temperature above the fluid saturation temperature. The major simplifications are that we treat the complex microlayer region macroscopically as an apparent three-phase contact line that continues the liquid-vapor interface, assumed always in local equilibrium, to the solid surface and works in a region of parameter space where conduction dominates convection. A very fast boundary integral method reduces the problem to a one-dimensional integral whose solution requires no trial-and-error procedure and allows for non-equilibrium contact angles. The calculation demonstrates that, for a solid far more conducting than the adjacent liquid, the overwhelming majority of the heat flux in the presence of a bubble on the surface occurs through the solid directly to the CL region, where it causes vaporization. This explains the jump in heat flux upon nucleation observed in experiments. A shift in parameters changes the predominant mechanism of heat transfer from rising bubbles to condensation at the bubble apex. The calculation finds a power law scaling of the heat flux with bubble density with exponents in the experimentally determined range and reproduces the single bubble subregime of standard heat flux vs temperature measurements. b. Quasistatic growth, deformation, and detachment of a vapor bubble in a liquid layer on a heating surface of finite thickness: Exploring the mechanism of nucleate boiling heat transfer-Part II. This portion of the work uses the above formulation to explore the quasi¬steady growth of a vapor bubble on the surface of a heated solid covered by a liquid layer, both of finite thickness, based on the corresponding quasistatic liquid and solid temperature distributions. The calculation assumes small Reynolds, Peclet and capillary numbers, which we check a postiori, and requires as an input knowledge of how the CL changes as the bubble grows, for which we propose three ad hoc models. Small bubbles grow as sections of spheres until buoyancy deforms them and causes them to neck. For two CL models, prior to necking, bubble volumes scale as t3/2 and slower after gravity causes necking until detachment. We assume slow bubble growth so that bubble shapes are at all times solutions of the Young Laplace equations that balance surface tension and buoyancy forces. A simple theory based on a mechanism whereby the CL acts as a strong heat sink for the heater explains these simulations and provides a scaling that collapses simulations for many parameter values onto a master curve. We show that our results are consistent with various experiments and with results of far more complicated theories. Upon detachment, necked bubbles can leave vapor residues on the solid surface whose volume strongly depends on the CL evolution input to the calculation. This residue has a CL, and thus still draws heat from the heater, yet the rising escaped bubble causes cooler liquid from above to flow downward, which can quench the residue. The fate of this residue may be relevant to the transition from the single bubble to the jets and columns subregions of nucleate boiling. As such, a better understanding of three-phase CL motion with evaporation is needed to clarify this transition. 2. Lattice Boltzmann approach. a. Before attacking the bubble growth issue, we first resolve certain numerical anomalies of the lattice Boltzmann/diffuse interface methods: Shrinkage of bubbles/drops in the lattice Boltzmann equation method for nonideal gases. One characteristic of multiphase lattice Boltzmann equation (LBE) methods is that the interfacial region has a finite (i.e., non-infinitesimal) thickness known as a diffuse interface. In simulations of, e.g., bubble/drop dynamics, for problems involving non-ideal gases, one frequently observes that the diffuse interface method produces a spontaneous, non-physical shrinkage of the bubble/drop radius. In this paper, we analyze in detail a single fluid two-phase model and use an LBE model for non-ideal gases in order to explain this fundamental problem. For simplicity, we only investigate the static bubble/droplet problem. We find that the method indeed produces a density shift, bubble/droplet shrinkage as well as a critical radius below which the bubble/droplet eventually vanishes. Assuming that the ratio between the interface thickness, D, and the initial bubble/droplet radius, r0, is small, we analytically show the existence of this density shift, bubble/droplet radius shrinkage and of the critical bubble/droplet survival radius. Numerical results confirm our analysis. We also consider droplets on a solid surface with different curvature, contact angle, and initial droplet volume. Numerical results show that the curvature, contact angle, and the initial droplet volume have an effect on this spontaneous shrinkage process. b. We have developed a Lattice Boltzmann method for the pressure tensor and chemical potential via the Helmholtz free energy to simulate the flow field and an entropy formulation of the energy equation for the temperature field for the solid-liquid-vapor system under gravity, both for the case of an infinitely thin solid as well as for a solid of finite thickness. Numerical agreement with similarity solutions to Stephan's and Scriven's problems validate our method. For the case of a finite thickness solid we can either solve for the temperature in the solid region by LB or by finite difference and the two solution methods agree well. The motivation for the LB approach is that, by treating liquid and vapor as a single phase of variable density, the CL dynamics do not need to be specified, but rather are an output of the calculation. In both solid cases we can follow bubble nucleation, growth, necking, and detachment and calculate detached and residue volumes. For infinitesimally thin solids, requiring the solid to have uniform temperature violates the presumption of our continuum model that CL is at the liquid-vapor saturation temperature. As such, these models cannot agree. Yet, the liquid-vapor interface far from the CL indeed turns out to coincide with the saturation temperature isotherm. Naturally, this coincidence is far better for the finite thickness solid case, where we indeed find an overwhelming heat flux through the solid to the CL. We calculate CL motion for both situations and compare with our earlier ad hoc CL motion models. We compare calculated heat flux vs solid superheat with data from measurements for the single bubble and jets-and-columns subregimes of nucleate boiling and look at how heat flux scales with system parameters. Bubble growth again follows V~t3/2 for the finite thickness solid case. 3. Molecular dynamics (MD). Since the physics of CL motion is poorly understood even in the absence of intense evaporation at the CL seen here, we step down to the molecular scale to resolve CL motion. As noted, we simulate the entire time course of this experiment from initial heating of the lower solid surface in a system otherwise of uniform temperature, through vapor bubble nucleation, growth and detachment. MD takes as its input intermolecular interactions and solves Newton's equations in time for many (O(10(6)-10(7)) particles to cull "macroscopic" behavior. Literature simulations of similar systems only nucleate bubbles by cavitation (negative liquid pressures). We reproduce such bubbles and observe their growth and detachment. More interestingly, we carry out constant pressure simulations (by a solid wall above the liquid that acts like a frictionless piston) and investigate various temperatures and pressures to come up with a phase diagram showing when stable bubbles nucleate, when only unstable vapor fluctuations occur and when no vapor is formed. We do this at first with no uniform body force (to mimic gravity). At this point, we introduce such a uniform body force. On the nanoscale, operative here, terrestrial gravity is utterly negligible. In order to mimic buoyancy on these scales we invoke a uniform body force in such a way as to yield the same Bond number (~0.3-1.0) as in a water-water vapor system at atmospheric pressure where bubble deform and detach. This force is ~10(10) times terrestrial gravity, but produces acceptable density differences in our system (<2%). This force makes it harder at positive pressure to nucleate bubbles. As such, we have nucleated bubbles at zero body force and, when they reach a critical size, we turn this force on and observe subsequent bubble evolution. This work is currently in progress, but requires additional funds for a student or portdoc to continue it. These publications are being prepared for submission to peer-reviewed journals: Lin Zheng, Taehun Lee, and David Rumschitzki "A thermal lattice Boltzmann study of nucleate boiling on a heated solid surface" to be submitted to Phys. Rev. E, March 2016. Jinyong Bao, Joel Koplik, and David S. Rumschitzki "Molecular dynamics study of the nucleation and growth of a vapor bubble in a liquid layer on a solid heated from below" In preparation for submission to Phys. Rev. E, estimated submission April 2016.

Bibliography Type:	Description: (Last Updated: 04/09/2020)
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